Design and Implementation of a Vulnerability Assessment Tool for Enterprise Linux Servers*

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Abstract

Cyberattacks caused by insufficient vulnerability management remain a major threat in enterprise environments. To address this issue, this study designed and implemented a Python- and YAML-based vulnerability assessment tool for enterprise Linux servers based on the KISA security inspection criteria. The tool focuses on administrative vulnerabilities related to account policies, permission settings, service access control, and log management, and supports both Python 2 and Python 3 environments. Experimental results showed that assessment outcomes may vary depending on the operating system version and default configuration. The tool supports vulnerability detection, remediation guidance, and re-assessment. Future work will expand inspection items and improve UI functions to enhance usability and practical applicability.

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1 Introduction

Cyberattacks that exploit vulnerabilities remain a major security threat, and large-scale data breaches have occurred due to insufficient vulnerability management [1]. These vulnerabilities can be classified into technical vulnerabilities, which arise from software and hardware assets, and administrative vulnerabilities, which occur when security policies are not properly reflected in system operations. In enterprise Linux server environments, weak configurations such as account policies, permission settings, service access control, and log management can lead to security incidents [2].

Linux servers are widely used to operate core infrastructures such as web services, databases, and authentication servers, and Ubuntu-based distributions are commonly adopted in enterprise environments due to their convenient package management, open-source ecosystem, and stable operation. However, insufficient security hardening during installation or operation may cause administrative vulnerabilities, making regular vulnerability assessments necessary. Therefore, this study designed and implemented a Python- and YAML-based tool to automatically assess administrative vulnerabilities in enterprise Linux servers based on the KISA security inspection criteria.

2 Vulnerability Management Tool

In this study, a vulnerability assessment tool compatible with Python 2 was developed for Ubuntu-based Linux server environments. Figure 1 shows the overall workflow of the proposed tool. The proposed tool manages security inspection criteria separately using YAML policy files. When the tool is executed, it parses the YAML policy files and extracts the inspection type, target files, required modules, option names, comparison operators, and reference values. The collection module then gathers the current system state from configuration files or command execution results. The decision engine compares the collected values with the policy criteria defined in YAML and classifies the results as PASS, FAIL, MANUAL, or ERROR. In addition, the tool provides decision evidence, collected evidence, error details, and remediation guidance to help security administrators understand vulnerabilities and perform follow-up actions. The inspection items were selected from 20 major areas that could lead to security incidents in enterprise Linux servers.

The tool was tested in Ubuntu 18.04.6 LTS and Ubuntu 22.04.5 LTS environments using Python 2 and Python 3. Ubuntu 18.04.6 LTS produced 12 PASS, 7 FAIL, and 1 MANUAL results, while Ubuntu 22.04.5 LTS produced 13 PASS and 7 FAIL results. After applying the remediation guidance and conducting re-assessment, some items changed from FAIL to PASS [4]. These results confirm that the proposed tool supports vulnerability detection, remediation guidance, and re-assessment.

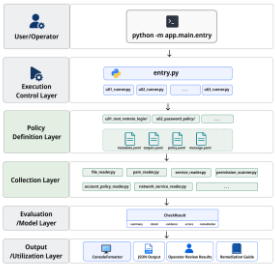


Figure 1 Vulnerability Management Tool Workflow

3 Conclusion

Vulnerability management is a key factor in preventing cyberattacks in enterprise environments. The vulnerability assessment tool proposed in this study supports the rapid and systematic identification of vulnerable elements that may occur in Linux server environments. In the first implementation phase, 20 important vulnerability assessment items for Linux servers were selected and applied from the server security items in the KISA security inspection criteria. In future work, the assessment scope and applicability of the tool will be expanded by adding more diverse inspection items, improving UI functions, and supporting various operating systems.

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